Emotion Recognition in Dance: A Novel Approach Using Laban Movement Analysis and Artificial Intelligence

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Abstract. Dance, as a highly expressive form of art, conveys intense emotions through bodily movements and postures. In the field of human-computer interaction, the automated recognition of dance movements poses a significant challenge concerning artistic expression and emotional classification. Analyzing dance movements enables us to extract rich emotional information. This paper introduces a novel approach for dance emotion recognition—the Laban Movement Analysis (LMA)—which characterizes the human body based on three aspects: body distribution, body structure, and dynamic trends. Leveraging artificial intelligence-based computer vision technology, we conduct a comparative analysis and supervised learning on existing dance performance video datasets. Various machine learning algorithms are trained and compared. The results indicate that recognizing emotional information from the perspective of dance movements achieves a high level of accuracy.

Keywords: Human emotion classification · Artificial intelligence · Laban movement analysis

1 Introduction

Human emotion recognition is currently a significant research direction within the realm of human-computer interaction, with widespread applications across multiple domains such as text, audio, and video [1]. Emotion recognition is a process that utilizes computer analysis to interpret various emotional information. This involves extracting features that describe emotions and creating a correspondence between these features and

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different emotional states. This further includes classifying emotions to infer an individual's emotional state. This process aims to comprehensively understand and interpret human emotions through computer technology. Meanwhile, affective computing, which focuses on the development of technology and theory to promote a profound comprehension of human emotions, has sparked extensive research initiatives and garnered notable scholarly attention. Affective computing is not merely a technical task but also an interdisciplinary field that concerns human-computer interaction, artificial intelligence, and psychology. Its research aims to better perceive, understand, and respond to human emotions through computer systems, providing a more intelligent and humanized experience for human-computer interaction. Among various human behaviors, human dance exhibits a clear expression of emotions. Dance, as one of the oldest art forms of humanity, is an important branch of ancient civilization. Its origin is rooted in labor and is interpreted through human body movements. Movements in dance have a unique power to vividly convey the thoughts and emotions of the dancer [2]. This makes human dance an extremely expressive art form, providing a rich and complex data source for emotion recognition research. While exploring the field of emotion recognition, special attention to the emotional expression of human dance will provide a beneficial perspective for a deeper understanding of the complexity and diversity of emotion recognition.

The Labanotation, invented by Rudolf Laban of Hungary, is one of the most widely used notation systems for recording human dance movements in the world today [3]. This method plays a pivotal role in areas such as choreography, dance training, and the documentation and analysis of human motion, demonstrating considerable potency [4–8]. Compared to other forms of recording such as video and images, Labanotation has the advantage of being preserved in paper or digital form, characterized by small storage space, good resistance to damage, and ease of preservation and dissemination. The elements of Labanotation include the basic orientation of the body, the quality of movement, and the direction of movement in space. Professional dance performers can easily read Labanotation, which facilitates the convenient recording, learning, and dissemination of dance. Therefore, as an effective tool for recording and inheriting dance, Labanotation plays a unique role in the preservation and dissemination of dance art.

Labanotation is an intuitive, clear, and scientific system of symbols for recording human movement, far superior to ordinary text and graphics in describing dance. By observing and analyzing human movement, it provides a framework for describing and understanding actions. Currently, the Laban method, as a description method for body movement, is extensively employed in emotion recognition [9]. Based on Laban Movement Analysis (LMA), researchers have studied the correlation between human movement and emotions. The results show that there is a good correlation between LMA features and emotions. Meanwhile, research on the automatic generation of Labanotation based on human movement analysis has made significant progress. However, how to recognize emotions in dance through its features still needs further study.

In this paper, we introduce a dance emotion recognition method based on Laban features (Laban Movement Analysis, LMA). The method identifies emotions by analyzing features in dance movement data, focusing primarily on the feature expression of body distribution, body structure, and movement trends. Through this approach, we aim to capture the rich expression of emotions in dance more intricately. Simultaneously, this

paper employs a multi-feature method, integrating these features into a neural network model to achieve automated emotion recognition, thereby avoiding the tedious process of manual emotion annotation. This neural network model, through learning a large amount of dance data, can abstract more complex emotional representations, improving the accuracy and universality of emotion recognition. Finally, to validate the effectiveness of our method, we not only developed a method for automatically extracting features and annotating dance emotions but also conducted comparative experiments with traditional manual emotion recognition methods. Through this series of experiments, we aim to demonstrate the superiority and feasibility of the proposed method in the field of dance emotion recognition. This has positive implications for promoting the development and application of dance emotion recognition technology.

2 Related Works

Labanotation [11], initially developed by Rudolf Laban, consists of two parts: a vertical staff and geometric symbols for dance. This Labanotation example spans three pages, read from bottom to top within a page and from left to right across pages, following the chronological order of the recorded dance movements. The body's midline divides the body into left and right sections. The staff primarily comprises nine vertical columns, each representing a part of the body. The columns on either side of the center line describe the most important parts of the body. From the center line outward, the columns sequentially represent the left and right support columns, left and right leg posture columns, torso movement column, left and right arm posture columns, and left and right hand movement columns. An additional head movement column is located on the far right.

The fundamental spatial symbols in the Labanotation system describe the horizontal and vertical directions of dance movements, which are crucial for the spatial analysis of Labanotation. The symbols for horizontal direction, with the central rectangle symbolize the "place" symbol. Centered on this symbol, other symbols represent "forward," "backward," "left," "right," "left-forward," "right-forward," "left-backward," and "right-backward," totaling nine symbols for horizontal direction. The symbols for vertical direction, sequentially from left to right as "low," "middle," and "high," represent by a different color. By combining the horizontal and vertical directions, space can be divided into 27 subspaces. Each subspace corresponds to a basic Laban symbol describing the direction of a dance movement. Therefore, there are 27 basic Laban symbols in the Labanotation system, representing 27 basic directions of dance body movements.

Aristidoue et al. [12] proposed a feature extraction framework for describing the four parameters of body, effort, shape, and space in Laban Movement Analysis (LMA). The Body component primarily involves the body structure and physical characteristics, describing the relationships, connections, and influences from other body parts or external factors in body movements. Furthermore, it reveals the sequence and relationships of body parts in motion, as well as general statements about body organization. The Effort component is dedicated to depicting the intention and dynamic characteristics of movements, including texture, sensation, and how energy is utilized in each movement. This part delves into the motivation, form, and energy distribution of movements, providing a

framework for a comprehensive understanding of emotional expression in motion. The Shape component focuses on how the body shape changes in motion. It describes the shape of the body in a static state, the relationship between the body and itself and the environment, and how the body changes towards a point in space. In addition, Shape also involves how the torso changes shape to support the movements of other body parts. The Space component describes the relationship between movement and the environment, path, and spatial tension lines. It details the evolution of movement in space, including interaction with the surrounding environment, path selection, and movement characteristics related to spatial tension lines.

With the rapid development of motion capture technology, Laban Movement Analysis (LMA) has been applied in multiple aspects, such as human body movements [9] and the relationship between human movement and emotions [9]. The Microsoft Kinect sensor has been used for Laban-based motion tracking and analysis [13–15]. Although the prices of motion sensors like Kinect are becoming increasingly affordable, they have not yet become mainstream for consumers. Conversely, the widespread adoption of video sharing and streaming platforms, from teleconferencing applications to social media, has solidified traditional video recording (sans additional sensors) as the principal method for capturing and documenting human actions. In professional contexts, such as clinical environments and performances, employing a single stationary camera has emerged as the prevailing norm for monitoring and recording human motion. Therefore, to better adapt to real scenarios, it is necessary to extract skeletal keypoint information from videos. BlazePose's pose estimation method [16], launched by Google Research, is a real-time pose estimation framework suitable for mobile devices. Figure 1 shows the effect of BlazePose on dance action pose estimation. During the inference process, the BlazePose model adopts a detector-tracker design, combined with the datasets used by BlazeFace, BlazePalm, and COCO, and proposes a new topology structure with 33 key points in the human body. With the rapid development of streaming media, recording dance through video has become mainstream. However, current methods are more or less limited to dynamic trends when it comes to the relationship between human movement and the emotions it represents, with inadequate consideration for static aspects.

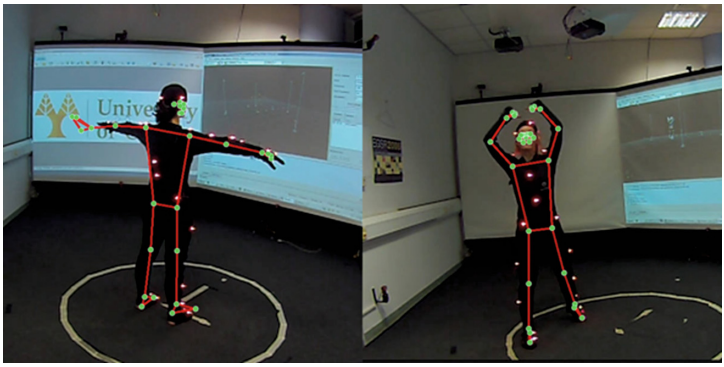

Fig. 1. BlazePose Key Points

Initially, we employ a series of artificial intelligence methods to process the original video clips of human dance movements. In the initial stage, we utilize human body detection algorithms and keypoint recognition methods to extract features of human skeletal keypoints. Subsequently, we mask redundant information to obtain keypoint distribution information and calculate feature vectors through intra-frame and inter-frame data for combination. Finally, we validate the effectiveness of this algorithm in emotion prediction through six algorithms (Decision Tree, Naïve Bayes, Support Vector Machine, Linear Model, Random Forest, Deep Neural Network). The uniqueness of this research method lies in the comprehensive application of multiple artificial intelligence technologies, from the extraction of human skeletal key points to feature vectors, and then to multi-algorithm verification, to improve the accurate prediction ability of emotions in dance movements.

3 Methods

Our methodological process aligns with most methods based on supervised learning and primarily includes the following four steps: First, we prepare and preprocess the dataset. In this stage, we are dedicated to organizing and processing data to ensure its suitability for model training. Second, we extract features related to body distribution, body structure, and dynamic trends. Through this step, we can capture key information to better describe the characteristics of the data. The third step involves training and testing the model using the processed data. In this phase, we employ supervised learning methods, allowing the model to learn the patterns in the data and evaluate its performance. The final step is to conduct experiments on entirely new data to verify the model's generalization ability on unseen data. Through these four processes, we can establish a comprehensive methodological framework to address our problem.

In our research, we utilize dance motion capture videos from the University of Cyprus [17]. This dataset consists of dance movement data from various performers. We employ the Laban Movement Analysis framework to obtain the required static and dynamic data features, necessitating the extraction of human skeletal data from the video data [12]. To identify the skeletal feature points in the video, we use MediaPipe [18], an open-source cross-platform framework based on graphic data processing, designed to build machine learning pipelines to handle streaming data, such as audio, video, and 3D point clouds. The MediaPipe skeletal keypoint detection module is an efficient and accurate human posture recognition algorithm that has been applied in fields such as human motion analysis, gesture recognition, and virtual fitting, recognizing human keypoints.

Figure 2 displays the information and total number of skeletal points extracted from the initial single frame. There are 33 points in total, including the nose, left eye (inner), left eye, left eye (outer), right eye (inner), right eye, right eye (outer), left ear, right ear, mouth (left), mouth (right), left shoulder, right shoulder, left elbow, right elbow, left wrist, right wrist, left pinky, right pinky, left index, right index, left thumb, right thumb, left hip, right hip, left knee, right knee, left ankle, right ankle, left heel, right heel, left foot index, and right foot index.

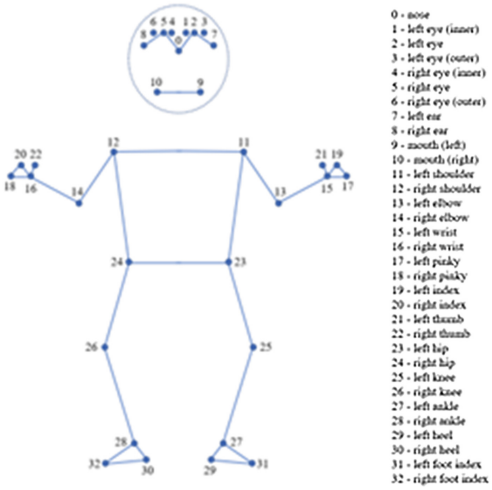

Fig. 2. MediaPipe Key Points

However, for static data, the information from 33 points is overly redundant. Therefore, we visually assess the motion trajectories of each point over a period of time based on the x and y coordinates of the video, thereby further filtering the key points to reduce subsequent computations. Ultimately, we obtain 13 points representing the body distribution of static data, as shown in Fig. 3, including the nose, left shoulder, right shoulder, left elbow, right elbow, left wrist, right wrist, left hip, right hip, left knee, right knee, left ankle, and right knee.

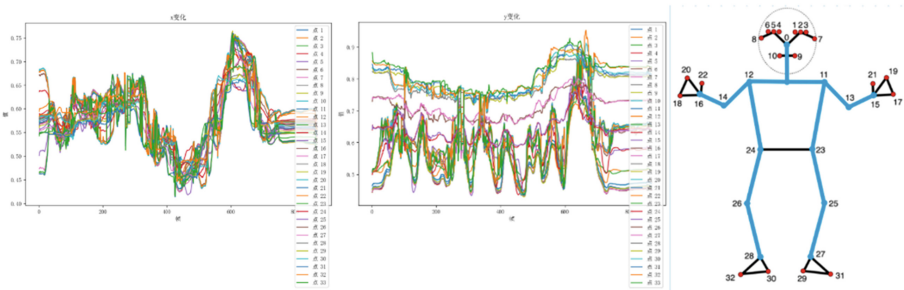

Fig. 3. Trajectory of Change

After obtaining the above data, namely the static data of body distribution, we calculate the features between frames, which include two parts: body structure and dynamic trends. The body structure includes the distance between skeleton points. In the present study, we undertake the computation of the relative distance between points on a skeleton. The feature equation, which quantifies the distance between a pair of skeleton points, is delineated in Eq. 1, where d represents the distance of the skeleton pair, and the coordinates of node i are (x_i, y_i), and the coordinates of node i's neighboring node j are

denoted as (x_j, y_j).

$$d = \sqrt{(x_i - x_j)^2 - (y_i - y_j)^2} \tag{1}$$

The trend of motion is a constituent element that correlates with the fluctuations in emotional states throughout the progression of the action sequence. The influence of force is an amalgamation of four distinct components: gravity, time, smoothness, and space. We calculated the speed and acceleration of the key points. Equation 2 shows the feature equation for the speed of the bone node, where v represents the speed of the skeleton point, the coordinates of node i at the current time are (x_i, y_i), and the coordinates at the previous moment are (x_{i-1}, y_{i-1}), t is the frame time. As delineated in Eq. 3, the feature equation for the acceleration of the skeletal point is presented. In this equation, a signifies the acceleration of the skeletal node, Δv denotes the change in velocity, and t represents the time duration of the frame.

$$v = \frac{\sqrt{(x_i - x_{i-1})^2 - (y_i - y_{i-1})^2}}{t} \tag{2}$$

$$a = \frac{\Delta v}{t} \tag{3}$$

Specifically, in this paper, with respect to the aspect of limb distance, we calculate the distance between the ankle joints and hip joints on both sides of the body, the wrist joints and the nose, and the wrist joints and the hip joints, as shown in Table 1. In terms of dynamic trends, we express them from two aspects: speed and acceleration. We learn the features of the wrist joints, hip joints, and ankle joints. At the same time, in order to combine static data with dynamic data, we define a window of 10 frames in length, and with the need for 3 window lengths, we can calculate all the required features. We calculate the data of three window lengths. In terms of body structure, we analyze the minimum standard deviation, obtain the most stable part within this section as the normalization factor, and normalize other structures. In terms of dynamic trends, we calculate speed and acceleration, and finally obtain dynamic data belonging to this section. After propagating it to the static data of its belonging frame, we obtain the static distribution within a frame and the dynamic information for a subsequent period of time.

Table 1. Body structure and dynamic trends

Distance	Velocity	Acceleration
Ankle and hip	Ankle	Ankle
Wrist and nose	Hip	Hip
Wrist and hip	Wrist	Wrist

In the end, we identify changes in dance emotions by inputting video sequences into a deep neural network and five machine learning models. Our dataset uses 16 videos

with 8016 frames as the training set, and 2 videos with 1002 frames as the test to verify the accuracy of the model. The distribution of the training set and the test set are shown in Table 2.

Table 2. Dataset

Emotion	Train set	Test set
Afraid	4008	501
Happy	4008	501

The five machine learning models we trained are Naïve Bayes, Support Vector Machines, Logistic regression, Decision tree and Random Forest. Naïve Bayes [19] can be applied to a range of tasks, such as classification and text classification. This algorithm has several variants, including Gaussian Naive Bayes, Bernoulli Naive Bayes, and Multinomial Naive Bayes. Support Vector Machines [20] is one of the supervised learning algorithms, which can be applied to classification, regression, and outlier detection tasks. The goal of this algorithm is to find a separable hyperplane in a high-dimensional space that can optimally divide the data into different categories or forecast the target value in a regression scenario. Logistic regression [21] is often used for binary classification tasks. It is an efficient and straightforward algorithm to execute, and it can yield successful outcomes in numerous real-world scenarios, especially when there is a roughly linear relationship between the target and independent variables. Decision tree [22] is a model that mirrors a tree-like structure and is employed for supervised learning and decision-making tasks. This algorithm divides the data into smaller, more manageable subsets according to certain criteria, with each division represented as a node in the tree. The attribute that offers the maximum information gain, quantifying the reduction in uncertainty or entropy of the target variable, is chosen for the divisions. Decision Trees can be used for both classification and regression tasks, and they are capable of handling both categorical and numerical attributes. Random forest [23] combines multiple decision trees to formulate a prediction. The algorithm is suitable for both classification and regression tasks. By combining individual trees, the model mitigates the variance and overfitting that can happen in a single decision tree, thus improving the model's stability and accuracy. Deep neural network is a type of feed forward neural network, including one input layer, one or several hidden layers, and one output layer. Information in the network travels unidirectionally, from the input to the output, without looping back. The data is processed at each layer and then sent forward to the next layer until the final output is derived. Every layer is made up of multiple neurons, with each neuron performing a fundamental computation on the weighted sum of its inputs to yield an output. The connection weights between neurons and each neuron's biases are established during the training phase using an optimization algorithm like gradient descent. In conclusion, our data feature extraction process and deep neural network model are illustrated in Fig. 4.

The evaluation of the performance was carried out using metrics such as precision, recall, accuracy, and F1 scores. Precision indicates the classifier's proficiency in correctly distinguishing negative samples, while recall reflects its capability to identify

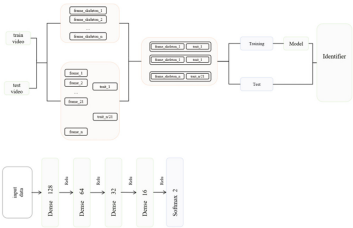

Fig. 4. Feature extraction and model structure

positive samples, with the highest possible score being 1. The F1 score, which is the harmonic mean of precision and recall, balances both metrics and ranges from 0 to 1. The corresponding formulas are as follows:

$$Precision = \frac{TP}{TP + FP} \tag{4}$$

$$Recall = \frac{TP}{TP + FN} \tag{5}$$

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{6}$$

TP = True Positive, TN = True Negative,
FP = False Positive, FN = False Negative

$$F1\,score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{7}$$

4 Results

In this paper, we compare the performance of five machine learning algorithms and a deep neural network on the same features. The results show that the deep neural network outperforms the other algorithms and achieves the highest accuracy, 78.94% on this feature set. Figure 5 presents the outcomes of six machine learning algorithms: Decision Tree, Naïve Bayes, Logistics Regression, Support Vector Machines, Random Forest, and Deep Neural Network.

Figure 6 depicts the confusion matrices for each machine learning method employed in this study. The diagonal elements represent correctly classified emotions, providing a visual comparison of model accuracy. Deep neural networks demonstrate a distinct advantage in terms of true positive (TP) rates compared to traditional models (Table 3).

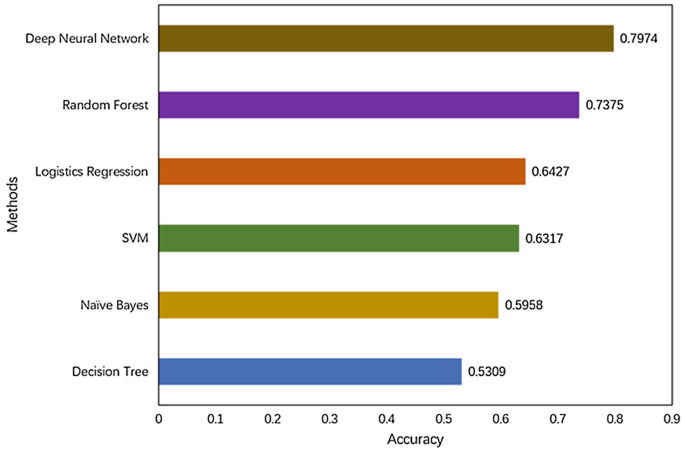

Fig. 5. Accuracy

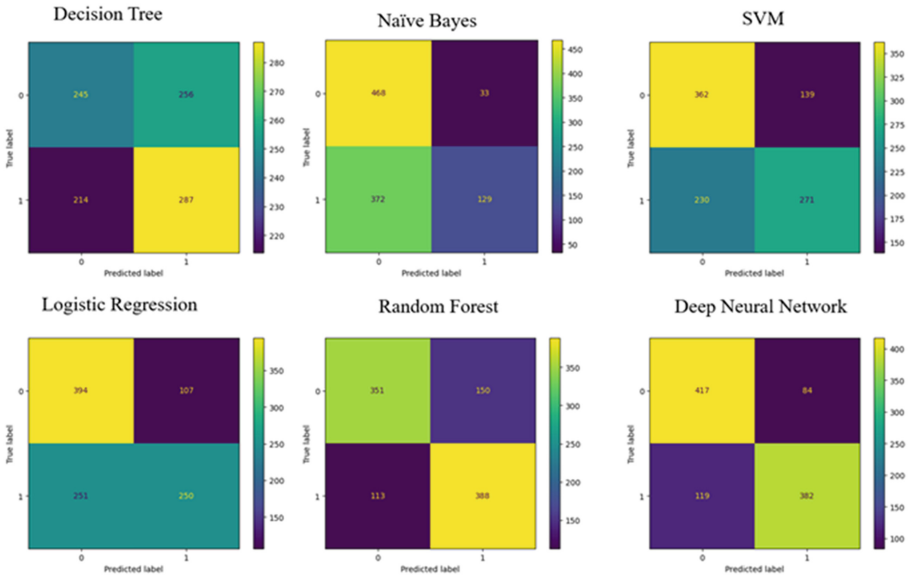

Fig. 6. Confusion matrix

Table 3. Emotion recognition using six machine learning algorithms

Algorithms	Precision	Recall	Accuracy	F1 score
Decision Tree	0.5285	0.5729	0.5309	0.5498
Naive Bayes	0.7963	0.2575	0.5958	0.3891
Support Vector Machines	0.6610	0.5409	0.6317	0.5950
Logistic Regression	0.7003	0.4990	0.6427	0.5828
Random Forest	0.7212	0.7745	0.7375	0.7469
Deep neural network	0.8197	0.7625	0.7974	0.7901

5 Discussion and Conclusion

In the present study, a novel approach was developed for the identification of emotions from video inputs, utilizing machine learning algorithms. The research encompassed the training and comparative analysis of six distinct machine learning algorithms, namely Decision Tree, Naïve Bayes, Support Vector Machines, Logistic Regression, Random Forest, and Deep Neural Network. These algorithms were trained using supervised learning techniques on pre-existing datasets of dance videos. Upon comparison of the six machine learning and deep learning methodologies, it was observed that the Deep Neural Network outperformed all other methods in terms of accuracy, achieving a rate of 79.74%. This was in contrast to the Decision Tree at 53.09%, Naïve Bayes at 59.58%, Support Vector Machines at 63.17%, Logistic Regression at 64.27%, and Random Forest at 73.75%, as depicted in Fig. 3. The findings of our study indicate that Deep Neural Networks exhibit superior performance in extracting high-level, hidden, and meaningful features from dance emotion classification.

Upon examination of the confusion matrix, it was observed that the deep neural network exhibited a total of 119 false negatives (instances where the true label was 1 but the predicted label was 0) and 84 false positives (instances where the true label was 0 but the predicted label was 1). The quantity of false negatives was found to be lower than those yielded by the other five machine learning methods under consideration. Furthermore, the Decision Tree method resulted in 214 false negatives and 256 false positives, while the Naïve Bayes method produced 372 false negatives and 33 false positives. Notably, these two methods not only had the highest rate of false classifications but also demonstrated an imbalance in their confusion matrices. The remaining three methods, namely Random Forest, Support Vector Machines, and Logistic Regression, exhibited relatively commendable performance on the confusion matrix for emotion recognition detection. Future work is warranted for a more in-depth analysis of the underlying mechanisms of these models.

Moreover, we conducted an in-depth analysis of the results. One of the primary causes of misclassification was found to be the reliance on the Mediapipe framework for keypoint extraction. In the context of video data, the capture of human keypoints may fail, leading to subsequent feature failure and erroneous emotion prediction outcomes. This suggests that the reliability of keypoint extraction poses a challenge when processing

video data. Therefore, more robust and reliable handling is required in this regard to enhance the accuracy of emotion prediction.

Despite the constraints posed by the limited data, our proposed Deep Neural Network exhibits a commendable level of accuracy in the detection of human motion for the recognition of emotional patterns, leveraging AI-based techniques. The outcomes of our research suggest the feasibility of automating the analysis of human emotions through the application of machine learning algorithms. These algorithms, which function on raw video inputs, could potentially pave the way for the development of AI-driven, standardized systems for the evaluation of human motion.

Our findings further imply that such AI-driven classification of movement could facilitate the establishment of fully automated reporting mechanisms in the domain of human emotion classification.

6 Future Work

The conveyance of emotion in dance is a multifaceted process, encompassing elements such as the performer, choreography, and environment. The task of identifying emotions in dance is notably challenging. This study seeks to address this complexity by refining the extraction of pertinent features, amalgamating them with deep neural networks, and evaluating the efficacy of this methodology through comparative experimentation. The success of dance emotion recognition hinges critically upon the richness and diversity of training datasets. Consequently, urgent research priorities lie in both enhancing the resilience of models to variations and expanding the existing body of dance emotion data.

The deep neural network model architecture utilized in this study is comparatively elementary. Prospective enhancements may be achievable by implementing more intricate models and integrating the forecasting of temporal data. This insinuates that the investigation of more elaborate architectures and the contemplation of temporal dynamics may offer potential for realizing superior outcomes in the domain of dance emotion recognition.

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